

GROUP B

Attempt any six Questions (6X5=30)

- ✓ 2. Write a program to find the factorial of a program using java? Take data from user [5]
3. Explain the operators available in Java Programming? [5]
4. What is OOP? Explain any four properties of OOP? [1=4]
- ✓ 5. What is a constructor? WAP a program to show parameterized constructor? [1+4]
6. WAP to find the factorial of a real number using recursion using java ? [5]
- ✓ 7. What is Inheritance? Explain different types of Inheritance? [1+4]
- ✓ 8. What is the difference between method overloading and overriding? Write the program for both ? [1=4]

GROUP C

Long Questions [2X10=20]

9. Explain final keyword in terms of variable, method and class? [5]
 - ✓ a. ? [1+4]
 - b. Write a complete program to show interface.
10. What is exception handling? Explain NullPointerException and Arithmetic Exception with program ? [1+4]
 - a. What is abstraction? How multiple inheritance is possible in java? [1+4]
11.
 - a. Explain "Final " Keyword. Explain the use of **Final** Keyword in **variable** and **method** and **class**
 - b. Write a program to use 'throw 'keyword in java. Show with example. [5]

B. Attempt any six questions 6X5=15 30

1. What is HTML5? How it is differ from HTML4?
2. What is an image map? Describe a key steps involved in creating a client image map in HTML using necessary tag and attributes?
3. What is container tag? Explain with suitable example?
4. Write an HTML code to print following output.

1. BCA

A. First Semester

- i. English
- ii. Mathematics
- iii. Society & Technology
- iv. Computer Fundamental
- v. Digital Logic

2. BBA

I. First Semester

- a. English
- b. Principles of Management
- c. Microeconomics
- d. Business mathematics
- e. Computer and Information Tec
App

5. Code your CV with pp size photo frame using HTML table Tag?

6. Write HTML code to prepare attractive Login form with 'OK' and Cancel button.

7. Write a process to imbedded google map in page?

C. Attempt any two questions 2x10=20

1. What is internal and external link explain with example?
2. What is frame? Explain with example?

*****Best of Luck*****

Saraswati Multiple Campus, Thamel
Bachelor in Computer Application
MID TERM ,2025

Course Title: DATA STRUCTURE AND ALGORITHM

Semester: 3RD

FM:60 PM:24

Short Questions: (6*5=30)

1. What is data structure. Explain its types with its importance?
2. Write an algorithm to convert infix expression to postfix expression?
3. What is circular queue ? Write an algorithm to insert an item in circular queue
4. What is AVL tree? Construct AVL tree of 50, 40, 35, 58, 48, 42, 60, 30, 33, 32
5. Convert the following expression from infix to postfix using stack:
 $(A + B) * (C * (D - E) + F) / G * (H - J)$
6. Create Huffman Tree and calculate What is double linked list? Write an algorithm to add a node at the beginning and end of double linked list?

Huffman code for the following character along with their frequencies using Huffman algorithm

Characters	A	E	I	O	U	S	T
Frequencies	10	15	12	3	4	13	1

Long Question: (2*10=20)

1. Introduce tree terminologies. Write algorithms to traverse binary tree using pre-order, post-order, and in-order traversal methods. Create B-Tree of order 5 using the following data: 6, 4, 22, 10, 2, 14, 3, 8, 11, 13, 5, 9 (2+3+5)
2. What is stack? List the application of stack. Write an algorithm to perform PUSH and POP operation?
3. What do you mean by MST? Explain Kruskal algorithm with example

Exam	Mid-Term		
Level	Bachelors	Full marks	60
Program	IT/A	Pass marks	24
Year/Part	III/I	Time	3 Hrs

Subject: System Analysis and Design

Batch: 2080

Group B: Short Answer Questions (6X5=30)

1. Explain the concept of an Information System. What are the different types of Information Systems, and how do they support organizational functions?
2. Describe the Systems Development Life Cycle (SDLC). What are the key phases involved in the SDLC?
3. What are the key differences between the Waterfall and Prototype SDLC models? Provide examples of when each model would be most appropriate.
4. What is CASE (Computer-Aided Software Engineering)? Explain how CASE tools support the system development process.
5. Discuss the concept of Rapid Application Development (RAD). What are its advantages and challenges compared to traditional development approaches?
6. What are the primary roles and responsibilities of a system analyst? How do system analysts interact with stakeholders during the development process?

Group C: Long Answer Questions (2X10=20)

1. Discuss the phases of the traditional Waterfall SDLC model. What are the strengths and weaknesses of using this approach in system development?
2. Explain the concept of Joint Application Development (JAD). What are its benefits in terms of gathering user requirements, and how does it differ from traditional methods of requirements gathering?

Thank You

Tribhuvan University
Faculty of Humanities & Social Sciences
Saraswati Multiple Campus 2025
Bachelor in Computer Applications

Course Title: Probability and Statistics
Mid-Term Examination

Code No: CAST 202

Time: 3 hours

Semester: III:

Candidates are required to answer the questions in their own words as far as possible

Group "A"

Attempt any SIX questions. [6*10= 60]

2. Describe the characteristics of measures of central tendency

3. Determine average wages from the following data:

Wages	25- 30	30- 35	35- 40	40- 45	45- 50	50- 55	55- 60	60- 65	65-70
No. of Worker	10	13	18	21	24	28	20	11	8

4. Calculate S.D. and compare the variability of two distributions

X: 5 6 3 2 6 7

Y: 3 5 8 4 3 2

5. The mean of two samples of sizes 50 and 100, respectively, are 54.1 and 50.3, and the S.D.s are 8 and 7.

Find the mean and S.D. of the sample size 150 obtained by combining the two samples.

6. An analysis of the weekly wages paid to the workers in two firms, A and B, belonging to the same industry, gives the following results:

	Firm A	Firm B
No. of workers	500	600
Average weekly wage (Rs)	586	575
Variance	81	100

i) In which firm is there a greater variability in individual wages?

ii) Calculate the mean of distribution of wages of all the workers of the firms A and B taken together.

7. Calculate the mean deviation from the median and its coefficient from the following data:

Age in years	No of persons	Age in years	No. of persons
10-20	15	50-60	15
20-30	14	60-70	11
30-40	16	70-80	9
40-50	20		

8. If the mode for the following distribution is Rs 24, find the mean deviation from the mean.

Expenditure in Rs	0-10	10-20	20-30	30-40	40-50
No. of families	14	23	27	?	15

Group "B"

Attempt any TWO questions.

[2*15 = 20]

Two samples describe students' ages in the regular daytime BCA program and the morning BCA program on campus. If class age homogeneity is a positive factor in learning, make a suggestion, with reason, which of the two groups will be easier to teach.

Regular BCA program or Morning BCA program

Regular Morning

Age 23 27 29 31 28 30 22 33 32 25

F 9 10 2 2 8 5 10 4 4 3

Morning BCA

Age: 40 35 38 33 32 34 36 37 39 31

F 2 4 1 5 7 3 4 5 1 6

10. A company paid bonuses to its employees as under:

Monthly Salary (Rs.)	Bonus (Rs.)	Paid	Monthly Salary (Rs.)	Bonus (Rs.)	paid
100-120	500		180-200	900	
120-140	600		200-220	1000	
140-160	700		220 and over	1100	
160-180	800				

The actual salaries of the employees were as given below:

Rs: 205, 190, 195, 218, 187, 168, 250, 168, 190, 168, 170, 175, 178, 175, 150, 125, 148, 165, 156, 145, 125, 110, 162, 130, 150, 184

Restate the data in the form of a frequency distribution and find out:

- The total bonus paid
- The average salary paid per employee
- The standard deviation of the distribution.

11. The time when a computer is out of operation due to breakdown or failure is called 'downtime'. The following data show the downtime in minutes of computer C for the last 70 working days:

Downtimes	Number of days
Never	10
Less than 10	24
Less than 20	44
Less than 30	54
Less than 40	62
Less than 60	68
Less than 120	70

(a) Find the arithmetic mean, standard deviation, and coefficient of variation of downtimes for computer C over the last 70 working days and interpret these results.

(b) If the mean and standard deviation of downtime of computer D over 70 working days are 18 minutes and 17 minutes, respectively, which computer is more reliable?